RaPD: Resolution-Agnostic Pixel Diffusion via Semantics-Enriched Implicit Representations

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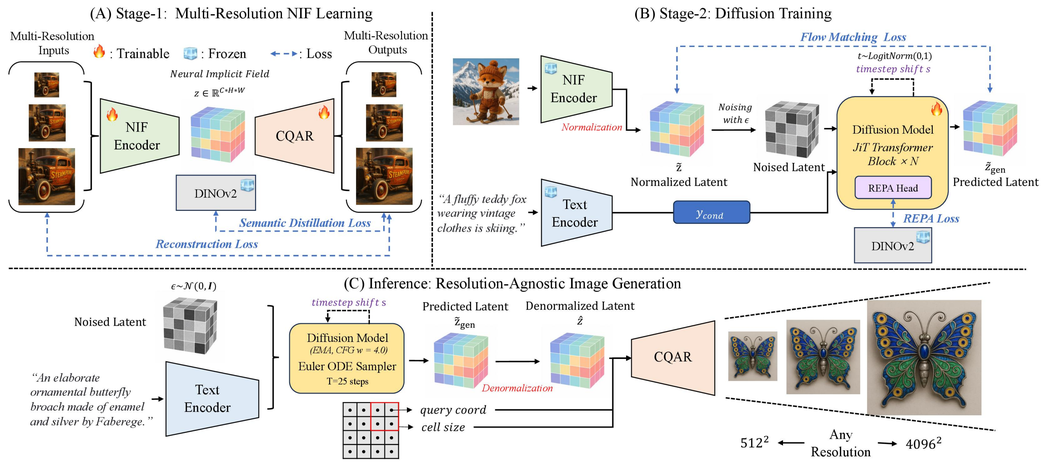
Can One Generated Representation Support Different Output Resolutions?

The visual world is continuous, but VAE-latent and pixel-token diffusion operate on fixed spatial grids. Higher resolution means more tokens and more diffusion compute.
Coordinate-based decoders support flexible output sizes, but reconstruction-trained latents are not necessarily ready for generation.



Our question:
Can we learn a generation-ready representation that renders at continuous coordinates?

Method: RaPD makes NIF generation-ready with SRG and CQAR



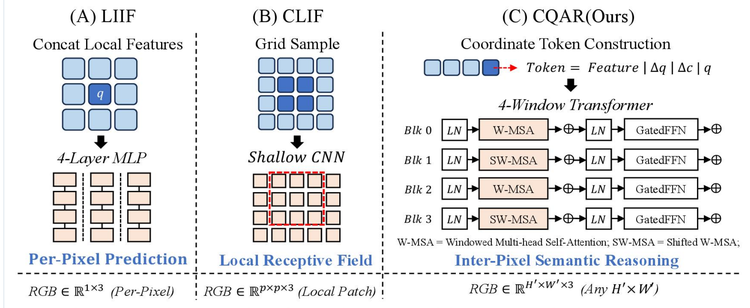
Stage-1: Learn semantically guided image-field representations.
Stage-2: Generate a fixed-size latent with text-conditioned flow matching.
Inference: Render one latent at different resolutions by changing the query grid.

SRG: Semantic Representation Guidance

- Semantic distillation. Pool and project the latent to align with frozen DINOv2 patch features, transferring semantic structure without changing the latent layout.
- Latent-density-aware timestep shift. Emphasize noisier states during training:
  $t = t_0 / [t_0 + (1 - t_0)s]$, $s = 32$

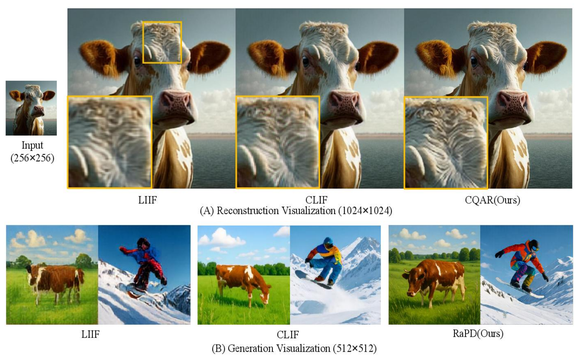
Make reconstruction-oriented latents generation-ready.

CQAR: Coordinate-Queried Attention Renderer



Inter-pixel reasoning for coherent, arbitrary-resolution rendering.

The Reconstruction–Generation Gap



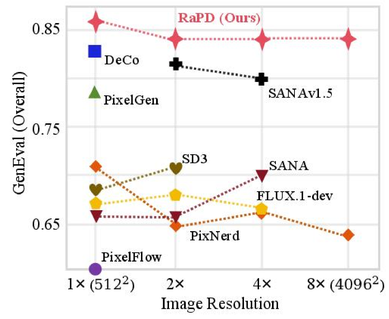
LIIF / CLIF: strong super-resolution reconstruction (top), but weak text-to-image generation (bottom).
Reconstruction training favors low-level texture rather than semantic structure.
Applying diffusion to these latents yields coarse semantics and poor visual fidelity.
Local decoders lack inter-pixel reasoning.

Arbitrary Resolution Generation Results



Training supervision reaches 1024²; rendering extends to 2048² and 4096². Only the query grid changes for a fixed generated representation.

Generation Quality and Performance



- **0.86 GenEval / 82.9 DPG**

Method	GenEval	DPG
PixNerd	0.71	81.5
PixelGen	0.79	80.4
DeCo	0.83	81.2
RaPD	**0.86**	**82.9**

- **Fixed diffusion cost**
- **4K in 2.13 seconds**

Method	512²	4096²
PixelGen	0.19 s	61.46 s
DeCo	0.21 s	62.72 s
RaPD	**0.23 s**	**2.13 s**

Why Semantic Representation Guidance Matters?

